Object Oriented Programming Lab (CSEG1044) Experiment-13&14: File I/O + Generics (Student Handout — Before Submission)

Tofik Ali

March 3, 2026

Contents

1 About this handout **1**

2 Experiment 13 (Syllabus) **1**

3 Experiment 14 (Syllabus) **1**

4 What you must do (minimum requirements) **2**
4.1 Task A — File I/O (Exp-13) 2
4.2 Task B — Generics basics (Exp-14) 2

5 Hints (read only if stuck) **2**

6 What to submit (MANDATORY) **2**

1 About this handout

This handout specifies the required programs for file handling and generics. Write and test your own implementation.

2 Experiment 13 (Syllabus)

- **Objective:** To perform file operations using Java I/O streams.
- **Problem Statement:** Develop programs using `FileReader` and `FileWriter`.
- **Expected Outcome:** Students will handle persistent data.

3 Experiment 14 (Syllabus)

- **Objective:** To implement generic classes and methods.
- **Problem Statement:** Design programs using generics.

- **Expected Outcome:** Students will understand type safety.

4 What you must do (minimum requirements)

4.1 Task A — File I/O (Exp-13)

Create `FileIOStreamsDemo.java`:

1. Write 3 lines to `input.txt` using `FileWriter`.
2. Read and print the file contents using `FileReader`.
3. Copy `input.txt` to `copy.txt` using a character buffer.
4. Print how many characters were copied.
5. Use try-with-resources and handle `IOException`.

4.2 Task B — Generics basics (Exp-14)

Create `GenericsDemo.java`:

1. Create a generic class `Box<T>` that stores a single value and returns it via `get()`.
2. Create a generic method `printTwice(T x)` and call it with different types.
3. Demonstrate at least:
   - `Box<Integer>`
   - `Box<String>`
4. Optional: create a generic `Pair<K,V>` and print it.

5 Hints (read only if stuck)

- Use try-with-resources to auto-close `Reader/Writer`.
- Relative file paths are resolved from your program's working directory (often your project folder).
- Generics prevent many `ClassCastException` issues by enforcing type checks at compile time.
- Avoid raw types (use `Box<String>` instead of `Box`).

6 What to submit (MANDATORY)

Submit a single zipped folder containing:

- `FileIOStreamsDemo.java`
- `GenericsDemo.java`
- sample output for both programs

- observations (5–10 lines): what try-with-resources does; where generics helped

Sources: Schildt, *Java: The Complete Reference* (12e) (nested/inner classes, exceptions, threads, I/O).; Oracle Java Tutorials: Nested Classes + Exceptions + Concurrency + Basic I/O (accessed March 3, 2026).; Java SE API docs (e.g., `Thread`, `AutoCloseable`) (accessed March 3, 2026). Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed March 3, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed March 3, 2026).